

Week 3 Internship Summary Report: Statistical Analysis

1. Executive Summary

This report presents the statistical analysis of two datasets: the cleaned Titanic passenger dataset and the cleaned Housing Price dataset. The analysis involved descriptive statistics, probability distribution analysis, hypothesis testing, and correlation analysis to identify patterns, relationships, and statistically significant differences within the data.

2. Dataset A: Titanic Passenger Manifest

Descriptive Statistics

Basic descriptive statistics were computed for the numerical and categorical variables. The average passenger age was approximately **29.14 years**, with a standard deviation of **13.49 years**, indicating moderate variation in passenger ages. The most common passenger profile was a **male** travelling in **Third Class** who embarked from **Southampton (S)**.

Probability Distributions

The **Age** variable showed an approximately bell-shaped (normal) distribution, while **Fare** was highly right-skewed due to a small number of passengers paying exceptionally high fares. Most passengers paid relatively low fares, with only a few purchasing premium tickets.

Hypothesis Testing

An independent two-sample t-test was performed to determine whether a significant difference existed between the average ages of male and female passengers.

- **Null Hypothesis (H_0):** There is no significant difference in the average ages of male and female passengers.
- **Alternative Hypothesis (H_1):** There is a significant difference in the average ages of male and female passengers.

The test produced a **p-value of 0.0021**, which is below the significance level of **0.05**. Therefore, the null hypothesis was rejected, indicating a statistically significant difference between the average ages of male and female passengers. On average, male passengers were older than female passengers.

Correlation vs. Causation

The correlation analysis showed a positive relationship between **Fare** and **Survived**. However, correlation does not imply causation. Paying a higher fare did not directly cause passengers to survive. Instead, higher fares were generally associated with higher passenger classes, which likely provided better access to lifeboats and contributed to the observed survival patterns.

3. Dataset B: Housing Price Market Metrics

Descriptive Statistics

The analysis showed an average house price of approximately **4,766,729**, with an average property area of **5,150 square feet**. The most common property configuration consisted of **three bedrooms**, was **semi-furnished**, and included **air conditioning**.

Probability Distributions

Both **Price** and **Area** exhibited positively skewed distributions, with skewness values of **1.21** and **1.32**, respectively. Most properties were concentrated within lower and middle price ranges, while a relatively small number of high-value properties created the long right tail.

Hypothesis Testing

An independent two-sample t-test was conducted to compare the average prices of houses with air conditioning and those without.

- **Null Hypothesis (H_0):** There is no significant difference in the average prices of houses with and without air conditioning.
- **Alternative Hypothesis (H_1):** There is a significant difference in the average prices of houses with and without air conditioning.

The test produced a **t-statistic of 10.6592** and a **p-value of 0.0000** ($p < 0.0001$). The null hypothesis was rejected, indicating that houses equipped with air conditioning have significantly higher average sale prices than those without.

Correlation vs. Causation

The correlation analysis identified strong positive relationships between **Price**, **Area**, and **Parking**. However, these relationships should not be interpreted as evidence of direct causation. Larger homes are more likely to include additional features such as parking and are often located in more desirable neighbourhoods. These related characteristics collectively contribute to higher property values.

4. Technical Conclusion

The statistical analyses of both datasets demonstrate the value of combining descriptive statistics, probability distributions, hypothesis testing, and correlation analysis to understand data. While correlation analysis helps identify meaningful relationships between variables, it cannot establish cause-and-effect relationships on its own. Reliable conclusions require statistical evidence together with an understanding of the underlying context of the data.